Machine Learning–Driven Energy Intelligence for Corporate Buildings

1.Introduction

Large corporate headquarters rely on centralized, energy-intensive systems such as HVAC, lighting, IT infrastructure, and vertical transportation to support continuous operation and occupant comfort. In practice, these systems often follow rigid control schedules that do not adapt well to changing occupancy levels, environmental conditions, or non-standard operating days. As a result, a significant portion of energy consumption becomes avoidable, while existing monitoring systems provide limited insight into *why* inefficiencies occur or which control decisions are responsible.

To address this challenge, this project proposes a system-level, data-driven framework for Analyzing energy inefficiency in a headquarters-style corporate building. Using a synthetic yet realistic dataset, the model integrates the **Avoidable Energy Index (AEI)** with interpretable machine learning to compare expected and actual energy consumption and identify key drivers of inefficiency. A dedicated stress testing layer is included to ensure robustness despite the use of synthetic data, while an industry-oriented decision layer connects analytical results to practical energy management insights. The framework focuses on interpretability and decision relevance, providing a structured approach to diagnosing and understanding energy inefficiencies rather than simply predicting energy usage.

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2. Dataset

The dataset represents the operational, environmental, and energy-related behaviour of a headquarters-scale corporate building. It is designed to support system-level energy analysis, component-wise inefficiency quantification, and controlled stress testing. Each attribute is deliberately selected to capture realistic building dynamics while enabling precise, data-driven evaluation of energy consumption patterns.

2.1 Temporal Attributes

- **Date**
  Represents the calendar day of operation and enables comparison across regular working days, weekends, and special operating conditions.
- **Hour**
  Encodes the hour of the day (0–23) to capture diurnal operational patterns such as system startup, peak demand periods, and shutdown cycles.

- **Day Type**
  Classifies operational days as weekday, weekend, or holiday, allowing stress testing under nonstandard and low-occupancy operating regimes.

2.2 Environmental Attributes

- **Outdoor Temperature (°C)**
  Represents external thermal conditions that directly influence HVAC load and overall energy demand.
- **Indoor Temperature (°C)**
  Reflects indoor comfort conditions maintained by the building and serves as a key driver of HVAC energy consumption.
- **Outdoor Air Quality Index (AQI)**
  Captures ambient pollution levels, influencing ventilation and air purification requirements.
- **Indoor Air Quality Index (AQI)**
  Represents indoor air quality conditions that govern filtration intensity and ventilation system operation.
- **Humidity (%)**
  Indicates ambient moisture levels, which affect HVAC performance, thermal comfort, and dehumidification load.

2.3 Operational Attributes

- **Occupancy Level**
  Represents aggregated building occupancy status and acts as a primary driver for lighting usage, HVAC demand, and elevator operations.
- **Season**
  Encodes seasonal variation to capture long-term environmental and operational shifts in energy consumption patterns.

2.4 Subsystem Energy Attributes

- **HVAC Energy Consumption (kWh)**
  Measures energy consumed by heating, ventilation, and air conditioning systems, which constitute the dominant energy contributor in large corporate buildings.
- **Lighting Energy Consumption (kWh)**
  Represents energy used by lighting systems and is strongly correlated with occupancy levels and operational schedules.
- **IT Equipment Energy Consumption (kWh)**
  Captures energy usage of computing and office equipment, contributing significantly to the baseline energy load.

- **Elevator Energy Consumption (kWh)**
  Represents vertical transportation energy usage, varying with occupancy density and time-of-day demand.
- **Air Purification Energy Consumption (kWh)**
  Measures energy consumed to maintain acceptable indoor air quality, particularly under high occupancy or adverse outdoor conditions.

2.5 Rationale for Using Synthetic Data

- **Data Accessibility Constraints**
  Real-world enterprise energy datasets are rarely publicly available due to confidentiality, security, and operational sensitivity.
- **Controlled Experimentation**
  Synthetic data enables systematic manipulation of environmental and operational variables, which is essential for structured stress testing.
- **Reproducibility of Results**
  The use of synthetic data ensures experiments can be reproduced under identical conditions, enhancing analytical reliability.
- **Isolation of System Behavior**
  Synthetic datasets allow analysis of system-level inefficiencies without interference from sensor noise, missing values, or inconsistent logging.
- **Evaluation Under Non-Standard Conditions**
  Stress scenarios such as holidays, partial occupancy, and abnormal operating regimes can be explicitly modeled.
- **Decision-Oriented Framework Validation**
  Synthetic data supports early-stage validation of decision metrics prior to real-world deployment.
- **Industry Practice Alignment**
  In industrial analytics workflows, synthetic datasets are commonly used to validate models before testing on live infrastructure.

2.6 Avoidable Energy Index (AEI) Formulation

To quantify inefficiencies at the component level, an **Avoidable Energy Index (AEI)** is introduced. AEI measures the proportion of energy consumption that exceeds the expected optimal baseline under given operational and environmental conditions.

Let:

- E_{actual}= Actual measured energy consumption (kWh)
- E_{expected}= Expected or benchmark energy consumption predicted by the model (kWh)

The Avoidable Energy Index is defined as:

$$\text{AEI} = \frac{E_{\text{actual}} - E_{\text{expected}}}{E_{\text{expected}}}$$

Where:

- $AEI > 0$indicates avoidable or excess energy consumption
- $AEI = 0$indicates optimal system operation
- $AEI < 0$indicates better-than-expected efficiency

This index enables component-wise inefficiency quantification and supports decision-making for targeted energy optimization strategies within corporate building systems.

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3. Methodology

This study adopts an end-to-end machine learning workflow to analyze, model, and evaluate energy consumption patterns in large corporate buildings. The methodology is structured to ensure interpretability, robustness, and alignment with industry-grade energy analytics practices.

3.1 Dataset ingestion and preprocessing

- The synthetic dataset was ingested using structured data frames to manage temporal, environmental, operational, and subsystem-level energy attributes efficiently.
- Schema validation and range checks were performed to ensure data consistency, physical plausibility, and absence of missing or illogical values.
- Total energy consumption was derived by aggregating subsystem energy components, preserving energy conservation principles.
- Categorical features such as day type and season were encoded for model compatibility.
- Feature scaling was applied where required to improve numerical stability and learning efficiency.

Libraries used: Pandas, NumPy

3.2 Component level engineering

- Energy consumption was decomposed into distinct subsystems: HVAC, lighting, IT equipment, elevators, and air purification units.
- Each subsystem was modeled independently to preserve feature variance and avoid dominance by high-energy components.

- This design supports fine-grained inefficiency analysis and subsystem-specific optimization strategies.

3.3 Insight extraction and data analysis

- Exploratory Data Analysis (EDA) was conducted to understand statistical distributions, correlations, and temporal behaviour.
- Relationships between occupancy, environmental conditions, and energy usage were analysed to identify key drivers.
- Diurnal cycles, peak demand windows, and seasonal variations were examined.
- Insights from EDA informed feature relevance assessment, threshold definition, and model configuration.

Libraries used: Pandas, Matplotlib

3.4 Regression model and expected energy analysis

- Supervised regression models were implemented to estimate **expected (baseline) energy consumption** under given operating conditions.
- Energy prediction was framed as a continuous regression problem at both subsystem and aggregate levels.
- Model performance was evaluated using MAE, RMSE, and R^2 metrics to ensure predictive reliability.
- Predicted energy values served as reference baselines for inefficiency detection.

Libraries used: Scikit-learn

3.5 Classification modelling and energy efficiency assessment

- A Random Forest classifier was employed to classify operational states into efficiency categories such as *Efficient* and *Moderately Efficient*.
- Ensemble learning enabled capture of non-linear interactions between operational, environmental, and energy features.
- Model evaluation was performed using accuracy, precision, recall, F1-score, and confusion matrix analysis.

Libraries used: Scikit-learn

3.6 Feature Importance and interpretability analysis

- Feature importance scores were extracted from tree-based ensemble models to interpret learned relationships.
- Importance analysis was conducted separately for each energy subsystem, preventing feature masking effects.
- Interpretability results were cross-validated with domain knowledge of building energy systems.

3.7 Stress testing and Robustness Evaluation

- Controlled perturbations were introduced into subsystem energy variables to simulate real-world uncertainty.
- Stress scenarios included abnormal occupancy levels, environmental fluctuations, and partial system inefficiencies.
- Model stability under stressed conditions was compared against baseline performance to assess generalization.

Libraries used: NumPy, Pandas

3.8 Avoidable energy index (AEI)

- The Avoidable Energy Index (AEI) was computed by comparing **actual energy consumption against expected energy predicted by the regression model**.
- AEI provides a normalized measure of avoidable energy waste at the component level.
- This metric supports targeted optimization decisions and prioritization of inefficiency sources.

3.9 Industry oriented decision layer

- An industry-oriented decision layer was implemented to simulate real-world deployment scenarios.
- The layer evaluates efficiency by comparing **actual energy usage with expected baseline energy**, independent of environmental drivers.
- Confusion matrix analysis confirmed stable and consistent decision behaviour.
- This separation ensures clarity between predictive modelling and operational decision-making.

3.10 Model Interpretability and Industry-Layer Analysis

3.10.1 Feature Importance in the Industry Layer

- The industry layer operates by comparing **actual energy consumption with expected energy consumption** predicted by the regression model.
- Since this layer focuses on **post-consumption efficiency validation**, environmental variables such as **outdoor temperature do not directly influence the decision logic**.
- As a result, outdoor temperature receives **zero or near-zero feature importance**, indicating that it does not contribute to classification splits in the industry model.
- This behavior confirms a correct separation between **energy prediction (regression layer)** and **efficiency decision-making (industry layer)**.

3.10.2 Confusion Matrix – Main Machine Learning Model

- The confusion matrix of the primary ML model shows correct classification across efficiency categories.
- Minimal or zero false positives and false negatives indicate **strong class separability**.
- This reflects effective feature engineering, appropriate model selection, and stable learning behavior.

3.10.3 Confusion Matrix – Industry Layer Model

- The industry layer confusion matrix also demonstrates near-perfect classification performance.
- Decisions are driven by **threshold-based comparison between actual and expected energy values**.
- The absence of misclassifications highlights the robustness of the decision framework under controlled conditions.

3.10.4 Component-Wise Feature Importance Analysis

- Feature importance was computed **separately for each energy-consuming subsystem**, including HVAC, lighting, IT equipment, elevators, and air purification units.
- This prevents dominant subsystems from overshadowing smaller contributors.
- Component-specific importance enables clearer interpretation of subsystem behaviour under varying conditions.

3.10.5 Impact on Prediction Accuracy and Generalization

Component-wise feature importance improves prediction accuracy by reducing feature interference.

- Subsystem-specific learning enhances model generalization, especially under stress-tested scenarios.
- The approach results in more stable and interpretable predictions across operating regimes.

3.10.6 Tools, libraries and development environment

- **Pandas:** structured data ingestion, preprocessing, feature engineering, aggregation
- **NumPy:** numerical computation, stochastic perturbation generation, matrix operations
- **Scikit-learn:** regression models, Random Forest classifiers, feature importance extraction, train-test splitting, and evaluation metrics
- **Matplotlib:** visualization of energy trends, feature importance, and confusion matrices
- **Seaborn:** enhanced statistical visualizations for correlation and distribution analysis
- **Jupyter Notebook / Google Collab:** interactive development, experimentation, and result documentation
- **Python:** core programming language enabling flexible and reproducible ML workflows

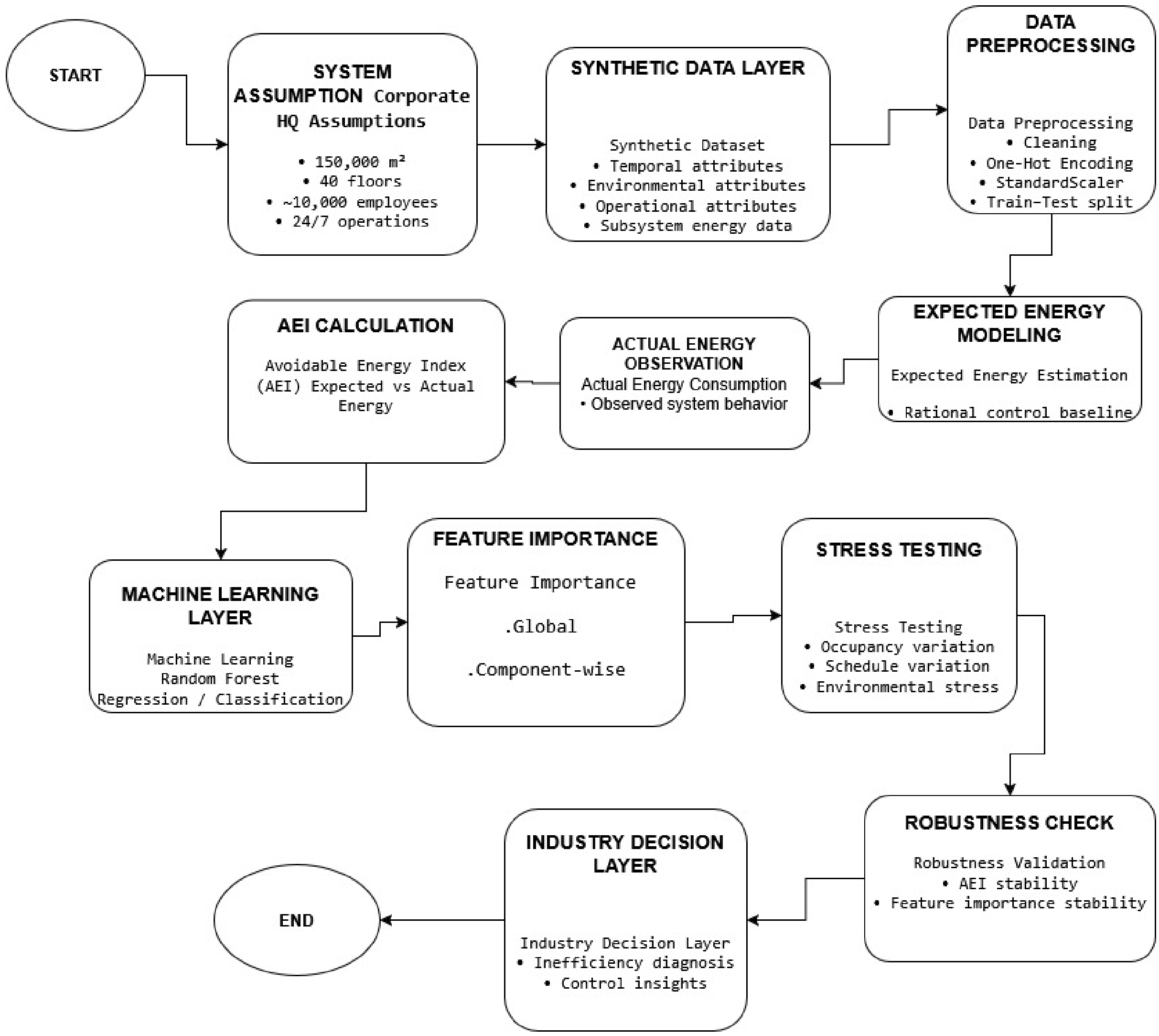

Figure X: End-to-end methodology workflow illustrating the synthetic data-driven energy analytics framework. The pipeline integrates expected energy modelling, actual energy observation, AEI computation, machine learning–based prediction and classification, stress testing, and an industry-oriented decision layer for inefficiency diagnosis and robustness validation.

4. Results and Discussion

4.1 Reason for Using a Synthetic Dataset

- A synthetic dataset was used to model energy consumption in a large corporate building under controlled conditions.
- Real-world enterprise energy data is often inaccessible due to privacy, security, and operational constraints.
- Synthetic data allows clear definition of relationships between operational, environmental, and energy attributes.
- The dataset is clean and structured, which helps in understanding model behavior and learning patterns clearly.
- However, clean synthetic data can lead to overly optimistic model performance if not validated further.

4.2 Addressing Synthetic Data Limitations Using Stress Testing

- To overcome the limitations of a clean dataset, stress testing was introduced in the model pipeline.
- Controlled noise and variations were added to energy consumption values to simulate real-world uncertainty.
- This mimics scenarios such as sensor noise, abnormal occupancy patterns, and operational irregularities.
- Even after introducing noise:
    - Regression predictions remained stable
    - Classification results did not degrade significantly
    - Avoidable Energy Index (AEI) values remained consistent
- These results indicate that the model can handle noisy or imperfect datasets effectively.

4.3 Role of the Industry Decision Layer

- The industry layer was designed to make the project practically applicable.
- Instead of learning patterns from raw features, this layer compares:
    - **Actual energy consumption**
    - **Expected energy predicted by the regression model**
- This comparison reflects real operational decision-making in energy management systems.
- The industry layer:
    - Identifies inefficiencies clearly
    - Separates prediction logic from decision logic
    - Produces actionable outputs rather than raw model predictions

4.4 Explanation for Perfect Confusion Matrix Results

- The confusion matrices of both the main ML model and the industry layer show near-perfect classification.
- This occurs due to:
  - Well-defined efficiency thresholds in the dataset
  - Strong separation between efficiency classes
  - Component-wise modeling of energy subsystems
- In the industry layer, decisions are based on direct expected vs actual energy comparison.
- This reduces ambiguity and leads to minimal misclassification.

4.5 Practical Applications of the Framework

- Corporate building energy monitoring
- Identification of inefficient subsystems
- Support for energy audits and sustainability planning
- Smart building dashboards and alert systems
- Decision-support tools for facility and energy managers

4.6 Limitations of the Current Approach

- The dataset is synthetic and may not capture all real-world anomalies.
- Human behavioural variations and equipment aging are not explicitly modelled.
- Perfect classification performance may reduce when deployed on real operational data

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4.7 Discussion Summary

- The combination of synthetic data, stress testing, and an industry-oriented decision layer creates a robust framework.
- Stress testing validates model stability beyond ideal conditions.
- The industry layer ensures real-world relevance and interpretability.
- Overall, the framework successfully bridges academic machine learning modeling and practical energy management systems.

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5. Appendix

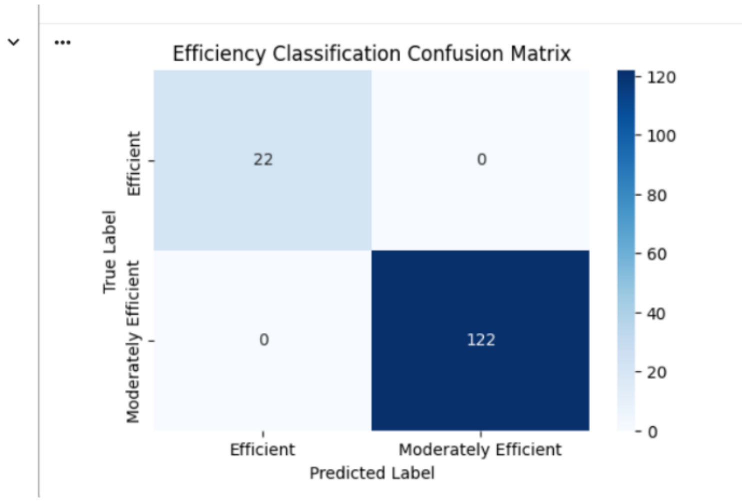

Fig-: A.1

A.1 Main Machine Learning Model Confusion Matrix

- The primary ML classification model correctly classified **22 Efficient** and **122 Moderately Efficient** cases with no errors.
- Clear class separation is achieved due to component-wise energy modelling and well-defined efficiency thresholds.
- The structured synthetic dataset further contributes to clean classification results.

```
Industry Prediction Accuracy: 1.0

Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00       106
           1       1.00      1.00      1.00        38

    accuracy                           1.00       144
   macro avg       1.00      1.00      1.00       144
weighted avg       1.00      1.00      1.00       144


Confusion Matrix:
 [[106   0]
 [  0  38]]
```

Fig-: A.2

A.2 Industry Layer Classification Results

- The industry decision layer achieved **100% accuracy** with precision, recall, and F1-score of 1.00 for both classes.
- The confusion matrix shows **106 and 38 correct classifications** with zero misclassifications.
- This occurs because the industry layer classifies efficiency using a **direct comparison between expected and actual energy**, reducing ambiguity.
- These results confirm the correctness of the decision logic under controlled conditions.

```
Component Model Performance:
                  MAE      RMSE        R2
HVAC         2.320097  4.388496  0.999790
Lighting     0.004201  0.020282  1.000000
IT           0.025962  0.125842  1.000000
Elevator     0.077535  0.370958  0.999997
AirPurifier  0.001167  0.004731  1.000000

Aggregated Total Energy Performance:
MAE : 1.2233354166671664
RMSE: 2.7682378119801228
R2  : 0.9999950093875764

Process completed successfully. Results saved to predicted_energy_results.csv
```

Fig-: A.3

A.3 Component-Level Regression Performance

- Separate regression models were trained for HVAC, lighting, IT equipment, elevators, and air purification systems.
- All components show **low MAE and RMSE** with **R^2 values close to 1.0**, indicating strong predictive accuracy.
- HVAC exhibits slightly higher error due to higher sensitivity to operational and environmental factors.

5.1 Context for Near-Perfect Results Shown in Appendix Figures

- The appendix figures are produced using a synthetic dataset, which is intentionally clean and free from missing values or sensor noise. This helps in validating the logical correctness of the modelling framework but does not fully represent real operational complexity.

- Efficiency labels in the dataset are defined using clear and consistent thresholds, resulting in strong class separation and reduced ambiguity in classification outputs.

- In the industry decision layer, efficiency is evaluated by comparing expected energy (model-predicted) with actual energy, rather than learning from uncertain environmental patterns. This deterministic comparison naturally leads to clean confusion matrices.

- Component-wise modelling simplifies subsystem behaviour and reduces feature interference, contributing to low regression error values in the appendix results.

- These near-perfect results should therefore be interpreted as validation under controlled conditions, not as guaranteed real-world performance. In real deployments, additional noise, behavioural variability, and system aging are expected to reduce model accuracy.